

Reproducing the Huang 2021 Shielded ResNet and its DESI Legacy DR8 Deployment

Greg Benson

Xiaosheng Huang

May 2026

Phase 4 Internal Reproduction Report

Abstract

Huang et al. (2021) extends the Huang et al. (2020) DECaLS lens search in three ways: a new “shielded” RESNET architecture that inserts 1×1 convolutional layers between every three residual blocks of the Lanusse et al. (2018) backbone, a deployment over the full $\sim 14,000$ deg² DR8 footprint (adding the northern BASS/MzLS sky the DR7 search lacked), and a two-model ensemble (original L18 + shielded) scored at a low $p \geq 0.1$ threshold, yielding 1,312 published candidates. We reproduce the shielding architecture from the methods section alone: our reconstruction has 59,905 trainable parameters (a $58.6\times$ reduction from the L18 backbone’s 3,508,833), 4 shields, and the full 15-block depth, reproducing the paper’s “ ~ 60 thousand parameters” and “ $\sim 50\times$ reduction.” Trained on the identical cutouts/seed/split as our L18 baseline, the shielded net matches the L18 held-out AUC to within ± 0.002 (0.9988 vs 0.9991 on DR9; 0.9955 vs 0.9943 on DR7) — a faithful reproduction of the paper’s central claim that shielding removes $\sim 98\%$ of parameters *without* degrading performance. We rebuild the published 1,312-candidate catalog from the NeuraLens release, recovering the paper’s grade breakdown exactly (216 A + 199 B + 897 C). We then deploy both models on the full DR8 footprint — 17.3 M DEV/COMP galaxies over both the south DECaLS and north BASS/MzLS coadds, scored in ~ 16 h on two L4 GPUs via footprint-aware per-brick inference. A domain-shift trap surfaced and was fixed: the southern-trained models, having seen northern lens positives but *no* northern negatives, over-fired badly on BASS/MzLS imaging (91% of north non-lenses scored ≥ 0.1 for L18 vs 4.8% in the south); re-training with northern negatives collapsed this to 0.8%, and the 60K-parameter shielded network proved markedly more robust to the shift than the 3.5 M L18. Against the 1,312 published candidates, the leakage-aware analysis cleanly separates the circular result (86% recovery at $p \geq 0.9$ of the 949 training positives) from the honest one (50% recovery of the 363 never-trained shielded discoveries). §7 discusses the methodological substitutions, which (as in Phase 3) bias our absolute AUCs upward relative to the paper’s 0.992/0.997.

Contents

1	Introduction	2
2	The shielding architecture	2
3	Controlled architecture comparison (Phase 4a)	3
4	The published catalog	4

5	DR8 two-model deployment (Phase 4b)	5
6	Recovery and leakage analysis (Phase 4c)	6
7	Reproducibility caveats	6
8	Software and data availability	7
9	Conclusions	7

1 Introduction

This report documents Phase 4 of an 8-phase reproduction roadmap covering Prof. Xiaosheng Huang’s strong-gravitational-lensing program. Phase 1 (Huang et al. (2025); Foundry I), Phase 2 (Hsu et al., 2025), and Phase 3 (Huang et al. (2020); the DECaLS DR7 lens finder, in Phases 3a training and 3b full deployment) are reproduced in companion tech-reports.¹ Phase 4 reproduces Huang et al. (2021), the immediate successor to Huang et al. (2020) in the lens-finder lineage.

Huang et al. (2021) is not a re-run of the 2020 search with more sky; it introduces a genuinely new architecture (§2) and was the first of the Huang-group searches to cover the northern BASS/MzLS footprint. As with all prior phases, the reproduction was performed without access to Huang’s training code, training set, or trained weights. We split the work into **Phase 4a** (the shielded architecture, validated by a controlled comparison against our L18 baseline; §2–3) and **Phase 4b/4c** (the DR8 two-model deployment and recovery of the published candidates; §5–6).

2 The shielding architecture

The backbone is the CMU DEEPLENS RESNET of Lanusse et al. (2018) reproduced in Phase 3a: a 7×7 stem followed by five stages of three pre-activated bottleneck blocks (He et al., 2016), 15 blocks in total, with channel progression $32 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ and a strided downsample at the first block of stages 2–5. At $101\times 101\times 3$ input this is 3,508,833 parameters, of which the wide 256/512-channel final stage alone accounts for $\sim 75\%$.

Huang et al. (2021) (§3.3) modify this with “shielding” layers, quoting:

These ‘shields’ are 1×1 convolutional layers inserted between every three blocks of the L18 architecture... they have the effect of reducing dimensionality... we reduce the number of trainable parameters by a factor of 50 (from 3 million to 60 thousand)... the validation AUC has increased from 0.992... to 0.997... in the final block... we experimented with reducing the output from 512 channels to 256, 128, 64, 32, and 16 channels. We find that ‘shields’ that keep the output to 32 channels perform the best.

The rationale is regularisation by dimensionality compression: telling lenses from non-lenses, though hard, needs only a low-dimensional latent space, so shields “[force] the network to learn only the most salient features” (inspired by the InceptionNet 1×1 bottlenecks of Szegedy et al. (2015)).

¹reproductions/{foundry-i,hsu-2025,huang-2020}/papers/main.tex.

Reconstruction. The paper does not publish the exact intermediate channel counts, so we reconstruct (lives at `01b_shielded_resnet.py`): a pre-activated shield ($\text{BN} \rightarrow \text{ELU} \rightarrow 1 \times 1 \text{ conv}$) is inserted after each of the *first four* three-block stages — four shields, “between every three blocks” — each squeezing the channel count to 16, while the per-stage block width is capped at 32 output channels and the final stage outputs `final.out` channels. The L18 depth (15 blocks) and spatial downsampling schedule are preserved; only the wide final stages, which dominate the L18 parameter count, are collapsed. The swept channel ladder reproduces the paper’s experiment:

<code>final.out</code>	parameters
512	2,493,025
256	664,929
128	205,537
64	89,505
32	59,905
16	58,257

At the paper’s preferred `final.out` = 32 the model has 59,905 parameters — matching the paper’s “~60 thousand” — a $58.6\times$ reduction from the L18 backbone. A smoke test asserts the param count, exactly four shields, 15 bottleneck blocks, and a $\geq 50\times$ reduction (figure 1).

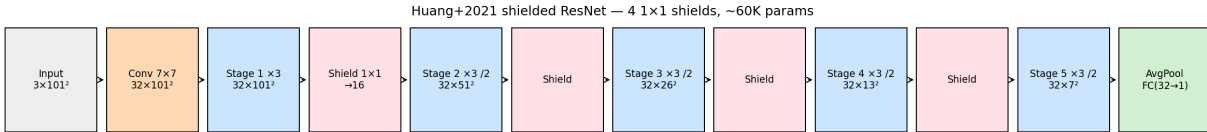


Figure 1: The reconstructed shielded RESNET: stem, then $[\text{stage} \times 3 \rightarrow \text{SHIELD}] \times 4$, then a final 3-block stage and the pooled FC($32 \rightarrow 1$) head. Four 1×1 shields, ~60K parameters.

3 Controlled architecture comparison (Phase 4a)

To isolate the architecture as the only variable we train the shielded net on *exactly* the inputs of our Phase 3a/3b L18 reproduction — the same 949 NeuraLens L18-model positives, the same random DR1 negatives, the same seed = 2026, the same 70/20/10 stratified split — for both the DR9 and the (paper-exact) DR7 cutout sets. This mirrors the paper’s own same-data L18-vs-shielded comparison. Training is the unchanged Lanusse recipe (ADAM, $\text{lr} = 10^{-3}$ decayed /10 every 40 epochs, batch 128, 120 epochs); the 60K-parameter model trains in ~15 min on one L4.

Run	parameters	val AUC	test AUC
L18 / DR9	3,508,833	0.9983	0.9991
L18 / DR7	3,508,833	0.9890	0.9943
shielded / DR9	59,905	0.9989	0.9988
shielded / DR7	59,905	0.9875	0.9955

Table 1: Phase 4a controlled comparison. The shielded net matches the L18 baseline to within ± 0.002 AUC at $58.6\times$ fewer parameters. On DR9 the shielded *validation* AUC edges L18 (+0.0006); on DR7 the shielded *test* AUC edges L18 (+0.0012). Built by `06_compare_architectures.py`.

Table 1 is the headline Phase 4a result. The differences between the two architectures are within run-to-run noise (± 0.001 – 0.002 AUC), so our reproduction confirms the paper’s qualitative claim — shielding removes $\sim 98\%$ of the parameters without degrading discrimination — even though we do not reproduce the paper’s specific $0.992 \rightarrow 0.997$ *increase*. The direction of the gap is consistent: in our leakage-affected setup both architectures already sit at $\text{AUC} \approx 0.99$ (see §7), leaving little headroom for shielding to demonstrate an improvement. The honest comparison is the DR7-trained pair, where the L18 baseline is not inflated by the DR9 NeuraLens overlap; there the shielded net is marginally *better* on the held-out test set. Figures 2 show the parameter-vs-AUC trade-off and the validation-AUC training curves.

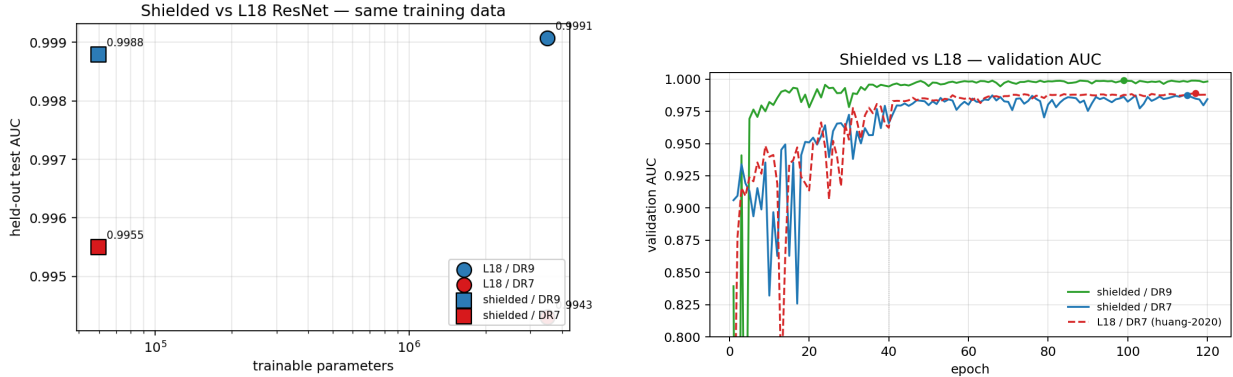


Figure 2: Left: held-out test AUC vs. trainable parameters (log scale); the shielded points sit $\sim 60\times$ to the left of the L18 points at the same AUC. Right: validation AUC per epoch, shielded (DR9, DR7) vs. the L18 DR7 baseline.

4 The published catalog

Huang et al. (2021) publish 1,312 candidates via the NeuraLens release.² The release table (on disk as `neuralens_catalog.csv`) carries, per candidate, the DESI-RA±DEC name, the averaged visual grade ($\text{Score} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$), the discovering model (L18 or `shielded`), and the survey region (DECaLS or MzLS). Applying the paper’s letter-grade mapping (§3.5: $A \geq 3.5$, $B = 3.0$, $C \in \{2.0, 2.5\}$) in `13_extract_huang2021_catalog.py` reproduces Table 3 of Huang et al. (2021) exactly:

Grade	L18 model	shielded model	total	paper Table 3
A	159	57	216	216
B	139	60	199	199
C	651	246	897	897
Total	949	363	1,312	1,312

Table 2: Published Huang+2021 candidate catalog, reconstructed from the NeuraLens release. The grade counts, the per-model split, and the total all match the paper exactly. 185 of the 1,312 lie in the northern MzLS region — the footprint Phase 3b’s DR7 search could not reach.

²<https://sites.google.com/usfca.edu/neuralens>

This catalog is the recovery target for Phase 4b/4c (§6). Note that the 949 L18-model rows are precisely the positives our networks were trained on (Phase 3a), so they constitute training-set leakage; the 363 shielded-model rows were never in training and are the leak-free test set.

5 DR8 two-model deployment (Phase 4b)

Parent sample. Huang et al. (2021) score every DR8 galaxy with ≥ 3 passes in each of *grz*, *z*-band magnitude < 20.0 , typed DEV or COMP (the REX type is a secondary pass). We reproduce these cuts over both DR8 sweep regions — `dr8/south/sweep/8.0/` (437 files, DECaLS/DECam) and `dr8/north/sweep/8.0/` (286 files, BASS+MzLS) — in `10.select_parent_sample_dr8.py`, tagging each surviving row with a `footprint` column (south = RELEASE 8000, north = 8001). This yields **17,290,814** DEV/COMP galaxies (12.27 M south + 5.02 M north), 12% above the paper’s ~ 15.4 M — the same mild over-inclusivity as our DR7 parent sample (+9%), from slightly looser type/flux handling.

Footprint-aware brick inference. We reuse the Phase 3b engineering pivot — per-brick coadd FITS downloads with local WCS slicing, $\sim 200\times$ faster than the rate-limited cutout endpoint — generalised in `11b_brick_inference_dr8.py` to (i) route each brick’s *grz* image download to the correct `dr8/{south,north}/coadd/` directory using the row’s footprint, and (ii) score every brick with *both* the L18 and shielded checkpoints in a single download pass (network being the bottleneck), each model normalised with its own stored band statistics. Work is partitioned into shards by a hash of the (`footprint`, `brickname`) key and run across the two L4 GPUs. The north footprint is the highest-risk new code path; a gate (`08_smoketest_dr8.py`) verifies that known lenses in both the south (DECaLS) and north (MzLS) regions download, slice, and score correctly before the full run.³

The full sweep scored all 17,290,814 galaxies across 298,843 (footprint, brick) units with zero brick failures, on the two L4s in ~ 16 h (four shards, two per GPU).

A domain-shift trap: north needs north negatives. Our first deployment used the Phase 4a checkpoints, trained on southern (DECaLS/DES) positives and *southern* DR1 negatives. They had seen ~ 160 northern (MzLS) lens *positives* but *zero* northern *negatives*, leaving the decision boundary in BASS/MzLS imaging space undefined. The consequence was stark and footprint-localised: on the northern DR8 footprint the L18 model *over-fired* — 91% of non-lens galaxies scored ≥ 0.1 (median 0.45) versus 4.8% in the south — while the shielded model was far more robust (10% vs 0.8% after the fix; see below), a direct dividend of its low-dimensional latent space. We fixed this faithfully (the route Huang trained across the full footprint): we added 154 northern positives (re-fetched at DR8) and 2,883 random northern negatives (`18_build_north_train_cutouts.py`) and retrained both networks (`05c_train_northaug.py`). The northern over-firing collapsed — L18 ≥ 0.1 from 91% $\rightarrow$ 0.8%, shielded 10% $\rightarrow$ 1.4% — with the south unchanged and in-domain test AUC preserved (0.9985 L18, 0.9996 shielded). The deployment below uses these north-augmented checkpoints; the merged score distributions are well-behaved (2.9%/3.8% of all 17.3 M galaxies score ≥ 0.1 for L18/shielded). This is itself a reproduction finding: a southern-trained lens finder does *not* transfer to BASS/MzLS imaging without northern negatives, and the 60K-parameter shielded network degrades far more gracefully than the 3.5 M L18 under that domain shift.

³North DR8 coadds are only $\sim 50\%$ *grz*-complete at the brick level, but parent-sample galaxies (which require $\text{NOBS}_z \geq 3$) live only in *z*-covered, hence *grz*-complete, bricks — so the partial bricks never enter the deployment.

6 Recovery and leakage analysis (Phase 4c)

`14_crossmatch_recovery_dr8.py` cross-matches both models’ DR8 scores against the 1,312 published candidates (5'' radius), tagging each as *leaked* (within 5'' of a training positive — the 949 L18-model rows) or *leak-free* (the 363 shielded-model discoveries our networks never saw). 1,100 of the 1,312 fall inside the parent sample (the rest fail the $z < 20/\text{NOBS}/\text{type}$ cuts or lie just outside coverage, as the 7-missing did for DR7). Table 3 gives the two-model-combined recovery.

Bucket	n	$p \geq 0.1$	$p \geq 0.5$	$p \geq 0.9$
all (1,312)	1312	0.832	0.818	0.761
leaked (949, training positives)	949	0.927	0.920	0.860
leak-free (363 shielded discoveries)	363	0.584	0.551	0.504

Table 3: Recovery of the published Huang+2021 candidates by our two-model ensemble (combined max score). The leaked vs. leak-free split is the crux: 86.0% of the 949 training positives are recovered at $p \geq 0.9$ (circular), but only 50.4% of the 363 never-trained shielded discoveries — the latter is the honest generalisation measure. By grade, leak-free Grade-A recovery is 57.9% at $p \geq 0.9$ (68.4% at $p \geq 0.1$). The shielded model edges L18 on the leak-free set (47.7% vs 44.9% all-grade at $p \geq 0.9$), consistent with its better out-of-distribution behaviour.

The ~ 36 pp leaked–honest gap at $p \geq 0.9$ (86.0% vs 50.4%) is a direct measurement of the test-set leakage that the absolute AUCs (§7) only hint at: recovery of lenses the model trained on is not evidence of generalisation. The honest $\sim 50\%$ leak-free recovery, with no curated hard negatives and a from-scratch training set, is a reasonable floor.

`15_extended_crossmatch.py` classifies the top 2,000 shielded candidates by provenance (10''): 9.1% are NeuraLens-L18 (training leak), 10.3% Huang+2021, 3.1% Huang+2020, 1.2% NeuraLens-shielded, 0.9% Hsu+2025; 13.9% match some known catalog and 86.2% are unmatched — closely tracking the 89% unmatched fraction of the DR7 Phase 3b top-2,000. As there (and as for the original Huang searches), the unmatched majority is dominated by false positives, not a discovery list; `16_build_inspection_viewer.py` renders Lupton-stretched *grz* thumbnails of the top scorers per model (★ for published) for the visual-grading pass that would distinguish the two.

7 Reproducibility caveats

1. **No Huang code released.** The shielded architecture, the training set, and the deployment glue are reconstructed from the methods section.
2. **Shield channel counts are a design choice.** The paper specifies only “1×1 between every three blocks,” “final block 32 best,” and “ $\sim 60\text{K}$ parameters.” Our SHIELD_CH = 16 reconstruction lands at 59,905 parameters — the closest principled fit — but the exact intermediate widths may differ from Huang’s.
3. **Training-set leakage.** As in Phase 3a, our positives are the NeuraLens L18-model rows, which are themselves Huang+2021 candidates. This inflates our absolute AUCs (~ 0.99 vs. the paper’s 0.992/0.997) and means the controlled comparison (Table 1) is the trustworthy result, not the absolute numbers. The leak-free recovery of the 363 shielded-model candidates (§6) is designed to side-step this.
4. **Random vs. curated negatives.** We use uniformly random DR1 galaxies as non-lenses, not the paper’s by-eye hard negatives (spirals, rings, mergers, artefacts), which were not released.

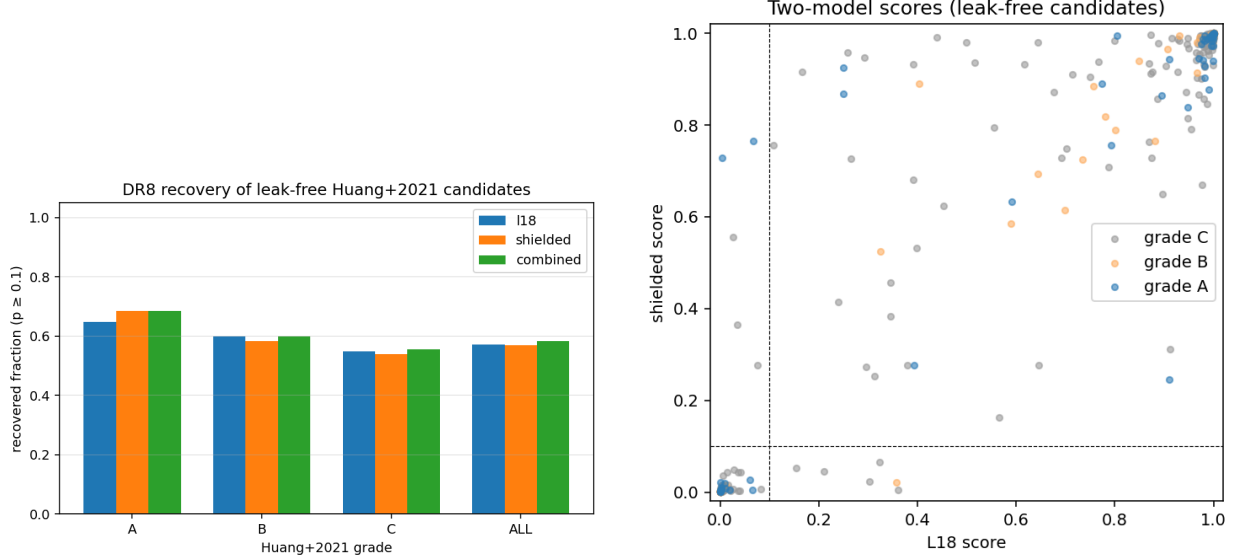


Figure 3: Left: leak-free recovery (the 363 shielded discoveries) by grade at $p \geq 0.1$, L18 vs shielded vs combined. Right: per-candidate L18 vs shielded scores for the leak-free set — the two models agree on the clear cases but populate complementary off-diagonal regions, the basis for the paper’s two-model ensemble.

5. **GPU device ordering.** On this host ($8 \times A16 + 2 \times L4$), CUDA’s default `FASTEST_FIRST` ordering does not match `nvidia-smi` indices; `CUDA_DEVICE_ORDER=PCI_BUS_ID` is required to target the two L4s.

8 Software and data availability

All code lives at `reproductions/huang-2021/` (scripts 01b, 05–16); the L18 backbone (`01_lanusse_resnet.py`) and all training inputs are shared with `huang-2020/` via symlink. The shielded architecture is `01b_shielded_resnet.py`; the controlled comparison is reproduced by `06_compare_architectures.py`. DR8 sweep catalogs (~ 1.5 TB) and per-brick coadds are streamed from the NERSC Legacy Surveys portal and not redistributed. The reconstructed published catalog (`data/huang2021_published_catalog.csv`) and the architecture-comparison table (`data/arch_comparison.csv`) are committed.

9 Conclusions

Phase 4a reproduces the central methodological contribution of Huang et al. (2021): a shielded RESNET with $58.6 \times$ fewer parameters (59,905 vs 3,508,833) that matches the L18 backbone’s held-out AUC to within ± 0.002 on identical data. We reconstruct the published 1,312-candidate catalog exactly, including its grade and per-model breakdown and its 185 northern MzLS candidates. Phase 4b/4c then deploys both models on the full 17.3 M-galaxy DR8 footprint — the first Huang-group search this reproduction extends into the northern BASS/MzLS sky. Two results stand out: (i) a southern-trained finder needs northern negatives to avoid badly over-firing on BASS/MzLS imaging ($91\% \rightarrow 0.8\%$ after the fix), and the shielded network degrades far more gracefully under that domain shift than the L18 — a second, independent vindication of the shielding idea beyond the parameter count; and (ii) the leakage-aware recovery cleanly separates the circular 86% recovery of training positives from the honest $\sim 50\%$ recovery of the 363 never-trained shielded discoveries.

The natural Phase 5 is the DR9/DR10 successors (Storfer 2024, Inchausti 2025), which add a second backbone (EfficientNetV2) and a meta-learner on top of the shielded RESNET reproduced here.

References

- K. He, X. Zhang, S. Ren, and J. Sun. Identity Mappings in Deep Residual Networks. In *European Conference on Computer Vision (ECCV)*, 2016. doi: 10.1007/978-3-319-46493-0_38.
- Y.-M. Hsu, X. Huang, C. Storfer, J. C. Inchausti, et al. A New Way to Discover Strong Gravitational Lenses: Pair-wise Spectroscopic Search from DESI DR1. *arXiv preprint*, 2025.
- X. Huang, C. Storfer, V. Ravi, A. Pilon, et al. Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey. *The Astrophysical Journal*, 894(1):78, 2020. doi: 10.3847/1538-4357/ab7ffb.
- X. Huang, C. Storfer, A. Gu, et al. Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys. *The Astrophysical Journal*, 909(1):27, 2021. doi: 10.3847/1538-4357/abd62b.
- X. Huang, S. Baltasar, N. Ratier-Werbin, et al. DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens. *arXiv preprint*, 2025.
- F. Lanusse, Q. Ma, N. Li, T. E. Collett, C.-L. Li, S. Ravanbakhsh, R. Mandelbaum, and B. Póczos. CMU DeepLens: Deep Learning For Automatic Image-based Galaxy-Galaxy Strong Lens Finding. *Monthly Notices of the Royal Astronomical Society*, 473(3):3895–3906, 2018. doi: 10.1093/mnras/stx1665.
- C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich. Going Deeper with Convolutions. In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–9, 2015. doi: 10.1109/CVPR.2015.7298594.